

The Motherboard Issue

The first run of the dual-core Athlon 64 X2 chips are compatible with any Socket 939/940 motherboard. All that's required is a BIOS update! But in the case of Intel, you need a new pair of supporting chipsets—the Intel 945P and 955X—meaning you'll have to buy a new motherboard if you want an Intel dual-core. This really makes it seem like AMD is already ahead in the race when it comes to dual-core.

Licensing issues

With the advent of dual-core, licensing issues have cropped up. If a software is licensed for one 'computer', but it's assumed that the computer has only one processor, what happens when the computer happens to have more than one processor? Does a dual-core processor count as two processors or one?

How multi-core chips will affect licensing is unclear, but industry analysts say a simple, uniform system is unlikely anytime soon. AMD recently released recommendations for how software companies should charge: it said companies should count processors based on the number of physical sockets used. This would mean a dual-core chip would count as a single processor. Microsoft, too, has said that

AMD could already be ahead in the race when it comes to dual-core

HyperTransport

HyperTransport is a high-speed technology for data transfer within the integrated circuits in computers and other devices. The technology allows data throughput in excess of 12.8 gigabytes per second. HyperTransport was developed by AMD in co-operation with several other companies, and is a trademark of the HyperTransport Consortium.

it will consider a dual-core processor a single processor, and will charge per processor, not per core.

Oracle, on the other hand, considers a dual-core server a two-processor server when charging for its server software. Similarly, IBM considers two cores as two processors for software licensing purposes. But other software sellers are taking the opposite stance: Sun not only prefers the socket definition for a processor, it argues for a radical pricing structure change. When selling its server and desktop software, it charges a fixed annual fee based on the customer's total number of employees.

Kuldip Hillyer, a manager in the strategic marketing group at BEA, gives a compelling argument: "Dual-core does not equal two CPUs in terms of performance. Hardware vendors don't charge you double; they charge a 30 to 40 per cent premium."

Dwight Davis, vice president and Practice Director at Summit Strategies, sums it up: "Some licensing models are showing signs of strain, none more than the per-CPU model common to databases and application servers. The static per-CPU model seems destined for extinction."

Here To Stay

Call it a game. Or a race. The chipmakers simply had to get faster CPUs out there, and dual-core seems the best approach—regardless of how much software support currently exists.

However, regardless of whether the software supports it, the fact is that more and more applications are being run these days at the same time. We routinely run anti-virus software, firewalls, CD-burning, Web browsing, and much more, all at the same time. If the OS can properly schedule these software to run on two cores, you *will* see the difference. In addition, software will evolve, and individual pieces of software will soon become amenable to being run on two cores.

Whatever the case may be, you're going to get a dual-core processor soon. They will be mainstream by 2006—there's no escaping that! ■

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HyperThreading

To understand HyperThreading (HT), one needs to first understand what threads are. Threads are something that programs create. Think about threads not as programs, but as independent execution lines. An example is seen in browser software: the graphics, text, sound, all seem to load independently—that's because each of these elements is a thread, which are interpreted by the browser as such.

Intel's HT technology basically fools the OS into thinking that there are two processors. If there had been two processors, it would have been possible to hand one thread to one processor and the other thread to the other one; with HT, that's what it seems like to the OS.

This ideally doubles the activity in the chip, and speeds it up. One example of how this speedup takes place is that with two threads being handed to the processor, there are fewer cache misses—a cache miss, as the term suggests, occurs when the processor requests something that isn't there in the cache.

Intel has estimated that a single instruction thread uses only about 35 per cent of a processor's available resources.

If there had been ideal division of labour between the two threads, they could have done something totally different—for example, one thread could operate on fetching data from the hard disk, and the other could do integer arithmetic. Of course, that's the ideal scenario, and the reality is that HT can increase the performance of some applications by "up to" 30 per cent, according to Intel.

The Pentium Extreme Edition 840 is a dual-core processor that supports HyperThreading within each core—so this means it appears as four logical processors to the OS!



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